

MADHUMITHA B

AI/ML Engineer • Python Developer • RAG & Computer Vision Enthusiast
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SUMMARY

Final-year B.Tech AI & Data Science student (CGPA 8.5) with 5+ months of industry experience at IITM Research Park. Builds end-to-end AI systems — RAG chatbots, multi-agent systems, computer vision pipelines & RPA automation. IEEE-published researcher and government copyright holder. Targeting full-time AI/ML or Python Developer roles at product-based companies.

EXPERIENCE

AI Engineer Intern | alfaTKG Co., Ltd. — IITM Research Park

Jan 2026 – Feb 2026

- RAG chatbot (FastAPI + Qdrant) with semantic chunking — 87% domain accuracy, ~30% latency reduction; steel defect detection (EfficientNet + PatchCore) — 94%+ anomaly accuracy.
- Automated 3 end-to-end industrial workflows via UiPath RPA — ~60% reduction in manual processing time.

AI/ML Intern | Plugzmart, Meras Plugins Pvt. Ltd. — IITM Research Park

Oct 2025 – Jan 2026

- EV charger performance forecasting model (89% accuracy) for predictive maintenance across 50+ station network; built EV domain chatbot with real-time charger status & diagnostic data.

PROJECTS

NeuroSphere AI — Multi-Agent RAG System — FastAPI • Next.js 14 • Groq LLaMA 3.3 70B • Qdrant • MongoDB Atlas • LangChain

- Production-ready ReAct agent (Think→Act→Observe): autonomous intent routing, semantic chunking + MMR reranking, per-user memory, voice I/O (STT/TTS); deployed on Render + Vercel with MongoDB Atlas, Qdrant Cloud & Cloudinary.

QueryVault RAG Chatbot — Python • FastAPI • Qdrant • MongoDB • LangChain

- End-to-end RAG pipeline — 87% domain accuracy, <1.5s response latency, validated on 200+ real queries.

Steel Surface Defect Detection — Python • EfficientNet • PatchCore • OpenCV

- Unsupervised anomaly detection — 94%+ accuracy, 20% false positive reduction via adaptive OpenCV preprocessing.

Dual Hand Sign Language Recognition — Python • MediaPipe • OpenCV • Flask

- Real-time dual-hand 26-sign ASL recognition — 91% accuracy, <100ms CPU inference, deployed via Flask UI.

TECHNICAL SKILLS

Languages: Python (Advanced), Java, C, HTML, CSS

AI/ML: TensorFlow, Scikit-learn, YOLOv8, EfficientNet, PatchCore, MediaPipe, OpenCV, Hugging Face

GenAI & NLP: LangChain, RAG Pipelines, Qdrant, FastAPI, Gemini API, Groq, LLaMA 3.3, Prompt Engineering

Tools & DB: UiPath Studio, Docker (basic), GitHub, MySQL, MongoDB, Power BI, Pandas, NumPy

EDUCATION

B.Tech — AI & Data Science | Rajalakshmi Institute of Technology, Chennai | CGPA: 8.5 / 10.0 2023 – 2027

Higher Secondary (Class XII) | E.V. Velu MHSS, Tiruvannamalai | 93.4%

2021 – 2023

ACHIEVEMENTS & RESEARCH

- IEEE Published (2025): First-authored research paper at IEEE International Conference.
- Govt Copyright SW-39797/2025-CO • School Pupil Leader — coordinated 200+ students • VP, Social Service Club (UBA) — led 50-member team, 10+ community outreach programs.
- Languages: Tamil (Native) • English (Fluent) • German (A1)

CERTIFICATIONS

Machine Learning Specialization — Coursera / DeepLearning.AI / Stanford | 2025

Data Analytics — NPTEL (IIT Certification) | 2025

MongoDB for Developers — MongoDB University | 2025